



Polifonia: a digital harmoniser for musical heritage knowledge, H2020

D4.3: Software for knowledge extraction from text – entities – 1st version (V1.0)

Deliverable information	
WP	WP4 Knowledge extraction from texts
Deliverable dissemination level	PU Public
Deliverable type	O Other
Lead beneficiary	OU
Contributors	OU, UNIBO, KNAW
Document status	Final
Document version	V1.0
Date	August 1, 2022
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Peer review	Philo van Kemenade (NISV) Abdul Shahid (NUIG)



This project has received funding from the European Union's Horizon 2020 research and innovation programme under grant agreement No 101004746

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Project Information

Project Start Date: 1st January 2021
Project Duration: 40 months
Project Website: <https://polifonia-project.eu>

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POLIFONIA Consortium

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2	OU	THE OPEN UNIVERSITY	United Kingdom
3	KCL	KING'S COLLEGE LONDON	United Kingdom
4	NUI GALWAY	NATIONAL UNIVERSITY OF IRELAND GALWAY	Ireland
5	MiC	MINISTERIO DELLA CULTURA	Italy
6	CNRS	CENTRE NATIONAL DE LA RECHERCHE SCIENTIFIQUE CNRS	France
	SORBONNE	SORBONNE UNIVERSITE (LinkedTP)	France
7	CNAM	CONSERVATOIRE NATIONAL DES ARTS ET METIERS	France
8	NISV	STICHTING NEDERLANDS INSTITUUT VOOR BEELD EN GELUID	Netherlands
9	KNAW	KONINKLIJKE NEDERLANDSE AKADEMIE VAN WETEN- SCHAPPEN	Netherlands
10	DP	DIGITAL PATHS	Italy

Project Summary

European musical heritage is a dynamic historical flow of experiences, leaving heterogeneous traces that are difficult to capture, connect, access, interpret, and valorise. Computing technologies have the potential to shed a light on this wealth of resources by extracting, materialising and linking new knowledge from heterogeneous sources, hence revealing facts and experiences from hidden voices of the past. Polifonia makes this happen by building novel ways of inspecting, representing, and interacting with digital content. Memory institutions, scholars, and citizens will be able to navigate, explore, and discover multiple perspectives and stories about European Musical Heritage.

Polifonia focuses on European Musical Heritage, intended as musical contents and artefacts - or music objects - (tunes, scores, melodies, notations, etc.) along with relevant knowledge about them such as: their links to tangible objects (theatres, conservatoires, churches, etc.), their cultural and historical contexts, opinions and stories told by people having diverse social and artistic roles (scholars, writers, students, intellectuals, musicians, politicians, journalists, etc), and facts expressed in different styles and disciplines (memoire, reportage, news, biographies, reviews), different languages (English, Italian, French, Spanish, and German), and across centuries.

The overall goal of the project is to realise an ecosystem of computational methods and tools supporting discovery, extraction, encoding, interlinking, classification, exploration of, and access to, musical heritage knowledge on the Web. An equally important objective is to demonstrate that these tools improve the state of the art of Social Science and Humanities (SSH) methodologies. Hence their development is guided by, and continuously intertwined with, experiments and validations performed in real-world settings, identified by musical heritage stakeholders (both belonging to the Consortium and external supporters) such as cultural institutes and collection owners, historians of music, anthropologists and ethnomusicologists, linguists, etc.

Executive Summary

Deliverable D4.3 is titled: Software for knowledge extraction from text – entities – 1st version. The deliverable summarises information about the software components under development in the Polifonia Work Package 4 dedicated to the identification and extraction of knowledge (entities) from textual resources. It is due month 18, and linked to Task 2: Automatic extraction of time, space, events, people and musical artefacts from text. The task leader is OU (The Open University) and the participants are: UNIBO, OU, and KNAW. This deliverable and task are strongly linked with research and development work undertaken in relation to pilots CHILD, MEETUPS, and MUSICBO, although the methods developed can be applied to other use cases as well. The resources described in this deliverable are released in the Polifonia Ecosystem. This is an interim deliverable, which will be followed by D4.4, due month 30.

Document History

Version	Release date	Summary of changes	Author(s) - Institution
v0.1	20/05/2022	Deliverable layout	Enrico Daga (OU)
v0.2	10/06/2022	Deliverable structured and subsections	Enrico Daga (OU)
v0.3	15/06/2022	Under development	Enrico Daga (OU) and Alba Morales (OU)
v0.4	25/06/2022	Sections 1,2, and 4 completed	Enrico Daga (OU) and Alba Morales (OU)
v0.4	25/06/2022	Section 3 completed	Arianna Graciotti (UNIBO)
v0.5	20/07/2022	Revised version incorporating feedback from reviewers.	Enrico Daga (OU), Alba Morales (OU), and Arianna Graciotti (UNIBO)
v0.9	20/07/2022	Release submitted to the coordinator.	Enrico Daga (OU)
v1.0	01/08/2022	Submission to EU	Coordinator

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1 Introduction

This deliverable collects the software components under development in the Polifonia Work Package 4 dedicated to the identification and extraction of knowledge (entities) from textual resources. The software presented here is the preliminary output of Task 2: Automatic extraction of time, space, events, people and musical artefacts from text. This deliverable and task are strongly linked with research and development work undertaken in relation to pilots CHILD, MUSICBO, and MEETUPS, although the methods developed can be applied to other use cases as well. The software components presented are:

- **Software for the extraction of historical musical meetups.** This component is composed of a set of sub-components, implementing a knowledge extraction pipeline. Although they together implement a composite process, from the textual biographies to a structured knowledge graph, the components can also be used separately, or customised for similar tasks.
- **Polifonia lexicon annotator.** The component provides means to annotate text with the polifonia lexicon, developed in Task 4.1.
- **Documentary evidence benchmark.** This components provides data and guidelines for evaluating the identification and extraction of listening experiences from text. It relies on the Listening Experience Database (LED) as gold standard.

The resources described in this deliverable are released in the Polifonia Ecosystem, either in version 1.0 (released in April 2022) and version 1.1 (released in July 2022). This is an interim deliverable, which will be followed by D4.4, due month 30.

2 Software for the Extraction of Historical Meetups

This section describes the components developed to extract information about Historical Meetups. We present a set of software tools for collecting, cleaning and analysing text from Wikipedia biographies description.

The components' releases are published on Zenodo [1, 2].

Background. This work is part of the MEETUPS pilot, which focuses on supporting music historians and teachers by providing a Web tool that enables the exploration and visualisation of encounters between people in the musical world in Europe from c.1800 to c.1945, relying on information extracted from public domain books such as biographies, memoirs and travel writing, and open-access databases.

Such encounters were particularly significant before the period when musical ideas and influences were widely disseminated through broadcasting and recording technologies, and the collection of data about them will make it possible to trace key points of cultural and musical exchange and dissemination and meetings that were catalysts for musical change. It may also reveal unexpected connections and relationships that cast new light on aspects of European music history. These encounters will be explored in a timeline and map interface and may reveal unexpected connections and relationships that cast new light on aspects of European music history. The tool will provide persistent, citable identifiers in order to support referencing in scholarship outputs.

Historical Meetups. A *historical meetup* has four components (see Figure 2.1):

- Participant(s): people involved (who?)
- Location: the place where it took place (where?)
- Encounter: the event that took place, the reason (what?)
- Time expression: the date or moment in time when the meetup happened (when?)

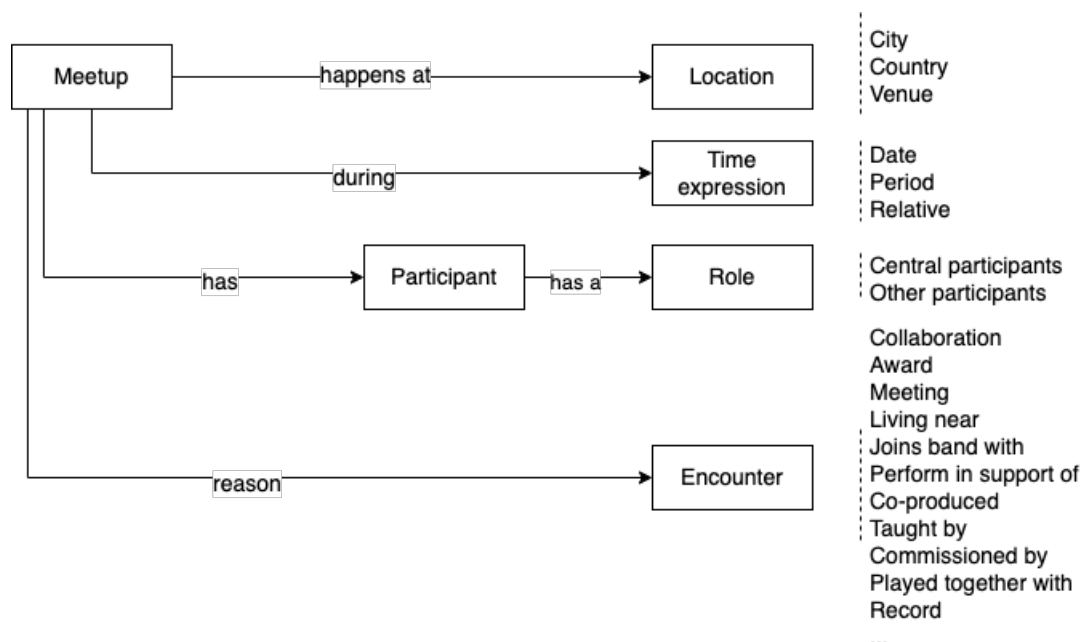


Figure 2.1: Historical Meetup definition

These requirements were compiled from interviews with scholars and Polifonia stories (Ortenz, David, and Sophia).

Knowledge graph construction pipeline. The information generated will be used to build a knowledge graph (KG) of historical meetups, which can be explored in a timeline and map interface by musical historians. In what follows, we describe the different components developed as part of the proposed KG construction pipeline (see Figure 2.2); here, we focus on the first three phases of the pipeline.

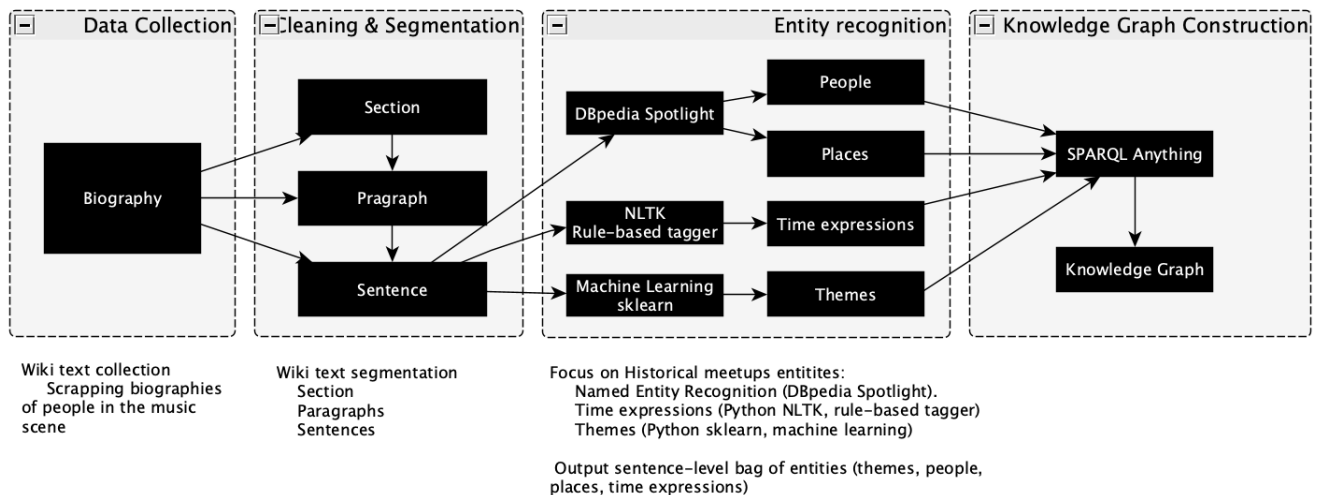


Figure 2.2: KG of Historical Meetups - construction pipeline

2.1 Corpus preparation: Data Collection

MEETUPS Corpus collection is a tool developed in Python. It collects Wikipedia web pages (in txt format) of European music authors.

2.1.1 Summary

ID	meetups-corpus-collection
Type	Software
Title	MEETUPS Corpus collection
Credits	Alba Morales (OU)
Description	MEETUPS Corpus collection is a tool developed in Python and PyCharm IDE. It collects Wikipedia web pages (in txt format) of European music authors. - Uses the Python library "wikipedia" to download Wikipedia webpages' text - Process the list of files in chunks of 100 units - The process can start and stop any time as it controls the last downloaded item
Source Code	https://github.com/polifonia-project/meetups_corpus_collection
Documentation	README.md
Release	v1.0
Licence	Apache 2.0
Running Instance	init.py
Related Deliverables	Polifonia deliverable D4.3

2.1.2 Exploitability

Meetups Corpus collection software contributes to the overall Polifonia goals; this software is part of the MEETUPS pilot and facilitates data collection from open-access websites such as Wikipedia. The data collected is the first step towards building a Knowledge Graph of Historical meetups; a total of 33,309 biographies are available for analysis.

2.1.3 Future work

An immediate application is using the corpus collected by the tool to extract historical meetups, part of the MEETUPS pilot.

2.2 Corpus preparation: Data Cleaning

MEETUPS data cleaning is a tool developed using Python and Jupyter Notebook. This software prepares the biographies content (collected as text files) in [meetups-corpus-collection](#) for the next step in the extraction of the historical meetups.

2.2.1 Summary

ID	meetups-data-cleaning
Type	Software
Title	MEETUPS data cleaning
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	MEETUPS Corpus data cleaning and segmentation is a tool developed using Python and Jupyter Notebook. It cleans and organises the corpus collected by "MEETUPS Corpus collection" available in https://github.com/polifonia-project/meetups_corpus_collection. The text collected from Wikipedia contains blank lines and sections that do not contain relevant information regarding historical meetups such as bibliography, references and discography. We use the Python library "spacy" to clean this data. Regarding the segmentation, data about a meetup is generally found in one sentence. Therefore, the tool splits the text into sentences. We complete the processing by adding indexes to each section, paragraph and sentence to identify the sequence in which text appears in the biography appear. The segmented data is then stored in CSV format. - Uses the Wikipedia authors' web pages collected by the MEETUPS corpus collection component - Clean blank lines, sections with no historical meetups data - Organise the text in sentences as the main unit to extract meetups information
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_data_cleaning.md
Release	v0.1
Licence	Apache 2.0

Running Instance	Jupyter Notebook 01_CleaningText.ipynb
Related Deliverables	Polifonia deliverable D4.3

2.2.2 Exploitability

This tool is part of the MEETUPS pilot and is used to clean and organise the text collected from Wikipedia. First, the text from Wikipedia contains line breaks and blank spaces between paragraphs that should be cleaned for later processing. The text also contains sections with information that does not refer to historical meetups. The tool removes all these data facilitating the segmentation of the text for processing.

Data about a historical meetup is generally found in one sentence. Therefore, the tool splits the text into sentences. We complete the processing by adding numerical indexes to each section, paragraph and sentence in order to identify the sequence in which each sentence appears in the biography. The final corpus is a collection of biographies organised by sentences, each one has an index that indicates the order in which appears in the biography. Each biography is stored using CSV format. The corpus will be exploited later to extract the different components of a historical meetup from each sentence in the biography.

2.2.3 Future work

Currently Meetups pilot is exploring the use of open-access web pages such as Wikipedia to obtain music personalities' biographies. If available, in the future, other types of data sources (e.g. books, memoirs) could be part of the pilot. Therefore, future work will expand the tool to include these data sources.

2.3 Identification of people and places

This component applies named entity recognition techniques for capturing people and places from biographies.

2.3.1 Summary

ID	meetups-entity-recognition
Type	Software
Title	MEETUPS identification of people and places
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This tool uses DBpedia Spotlight to identify and annotate possible entity mentions from input text. This is an essential process to identify two of the three main elements that define a meetup: people (who participated) and place (where). It uses as input the corpus of music personalities generated by the MEETUPS cleaning component https://github.com/polifonia-project/meetups_pilot/blob/main/01_CleaningText.ipynb. The implementation is divided into two Jupyter notebooks: The first notebook is in charge of querying DBpedia Spotlight, processing the responses (JSON format) and storing responses locally. The second notebook uses the responses from DBpedia Spotlight to capture data about people (identified as DBpedia:Person) and places (identified as DBpedia:Place).
Source Code	https://github.com/polifonia-project/meetups_pilot

Documentation	README_people_places_identification.md
Release	v0.1
Licence	Apache 2.0
Running Instance	Jupyter Notebook 02_queryDbpedia.ipynb 02_Identify_PP.ipynb
Related Deliverables	Polifonia deliverable D4.3

2.3.2 Exploitability

This tool is part of the KG construction pipeline. The entity recognition task is in charge of identifying and extracting two out of four components of a historical meetup: participants or people involved (who) and places (where). The data generated by this component will be used to build a knowledge graph of historical meetups.

2.3.3 Future work

Future work includes the annotation of the entire corpus collected by "MEETUPS data cleaning" component.

2.4 Identification of temporal knowledge

2.4.1 Summary

ID	meetups-time-extraction
Type	Software
Title	MEETUPS - Identification of temporal knowledge
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find time expressions. This is part of the process to identify one out of the four elements that define a meetup: the date or moment in time when it happened (when), along with data of the people involved (who), the place (where) and the event that took place (what) complete the historical meetup information. This implementation is a rule-based Time Expression recognition tagger based on research by [3] and SynTime software (https://github.com/zhongxiaoshi/syntime). The authors implement a three layer system that recognises time expressions using syntactic token types and general heuristic rules. Unlike SynTime, our software uses NLTK Toolkit and was developed using Python. Furthermore, this component classifies time expressions into three types of expressions: time range (e.g., from XX to XX, from XX, to XX, until XX), time point (e.g., exact date, 23/03/1294), and time reference (e.g., usually incomplete dates (19 April), two weeks, later this year).
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_time_expressions.md
Release	v0.1
Licence	Apache 2.0
Running Instance	Jupyter Notebook 03_Identify_TimeE.ipynb

Related Deliverables	Polifonia deliverable D4.3
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2.4.2 Exploitability

This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find time expressions. Time expressions, along with data of the people involved (who), the place (where) and the event that took place (what), complete the historical meetup information. The information generated will be used to build a knowledge graph of historical meetups.

Unlike SynTime, our tool classifies time expressions into one of the three types of expressions: time range (e.g., from XX to XX, from XX, to XX, until XX), time point (e.g., exact date, 23/03/1294), and time reference (e.g., usually incomplete dates (19 April), two weeks, later this year).

2.4.3 Future work

Future work includes evaluating the accuracy of the extracted time expressions. We consider it essential to compare the tool performance against state-of-the-art methods such as SynTime on well-known datasets such as TimeBank and WikiWars.

2.5 Identification of themes

This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find the type of event. MEETUPS identification of themes is a tool developed using Python and Jupyter Notebook, Python SKLEARN and a set of Machine Learning algorithms to classify sentences according to the established type of events.

2.5.1 Summary

ID	meetups-themes
Type	Software
Title	MEETUPS - Identification of themes
Credits	Enrico Daga (OU)
Description	This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find encounter types. It uses "sklearn" and a set of Machine Learning algorithms to classify sentences according to the established type of events. The tool extracts information from one of the four elements defining a meetup: the type of encounter (what). Encounter type, along with data of the people involved (who), the place (where) and the time it took place (what), complete the historical meetup information.
Source Code	https://github.com/polifonia-project/meetups_pilot
Documentation	README_identification_themes.md
Release	v0.1
Licence	Apache 2.0
Running Instance	Jupyter Notebook MeetupType_applyClassifier.ipynb
Related Deliverables	Polifonia deliverable D4.3

2.5.2 Exploitability

This tool is part of the MEETUPS pilot and processes text from music personalities' biographies to find a description of the encounter type. Encounter type, along with data of the people involved (who), the place (where) and the time it took place (what), complete the historical meetup information.

The information generated will be used to build a knowledge graph of historical meetups, which can be explored in a timeline and map interface by musical historians.

2.5.3 Future work

Future work includes using the encounter types annotations in combination with data generated by other components to generate the knowledge graph of historical meetups.

3 Polifonia Music Lexicon Annotator

The Polifonia Music Lexicon Annotator [4] is a software that allows to identify musical concepts in text. This module takes a text as input and annotates the content words in it with word sense information. Word senses are taken from WordNet [5], employing the Polifonia Lexicon, developed within the project and released with Deliverable 4.1 [6], to identify if a particular word sense is related to music or not. The disambiguation algorithm used to associate word senses with word form is based on the most frequent sense heuristic, computed on the SemCor dataset [7]. The output annotation is in JSON and includes tokenization, lemmatization and part of speech tagging, word senses and a flag that indicates if a particular concept in a sentence is related to music.

ID	polifonia-musical-lexicon-annotator
Type	Software
Title	Polifonia Musical Lexicon Annotator
Credits	Rocco Tripodi (UNIBO) Arianna Graciotti (UNIBO)
Description	This software provide an easy to use tool for recovering musical concepts in a text, taking the Polifonia Lexicon, based on WordNet as a reference
Source Code	https://github.com/polifonia-project/MusicalLexiconAnnotator
Documentation	README.md
Release	v0.0.1
Licence	CC-BY
Bibliography	Rocco Tripodi et. al, "D4.1: Plurilingual corpora containing source texts in English, French, Spanish and German (v1.0)" (2021).
Related Deliverables	Polifonia deliverable D4.3

3.0.1 Exploitability

This module lands itself very well for a fast annotation of textual documents. It allows to identify in a large corpus only documents that contain key-concepts relate to music. This feature makes it very useful for the study of musical terminology or for the selection of documents relevant to certain aspect of music.

3.0.2 Future Work

We plan to integrate into the repository different disambiguation algorithms in order to have better performance especially on the music domain. In fact, the main limitation of this module is that if the disambiguation is wrong also the identification the resulting music concept will be wrong. We plan also to extend the language coverage, including all the languages covered by the Polifonia Textual Corpus, i.e., Dutch, English, German, French, Italian and Spanish.

4 Documentary Evidence Benchmark

This software and data provide the benchmark for the extraction of documentary evidence, taking the Listening Experience Database (LED) as a reference. Motivation for this benchmark is outlined in this publications [8, 9]. The component release is published on Zenodo [10].

Below is information on the tasks, data, and the process that generated them from the LED database, also included (see file led-SNAPSHOT.nt.tar.gz).

The sources used to generate the benchmark are:

- "CHILD exemplar sources.xlsx": A curated list of sources from LED that also include experiences relevant to childhood
- corpus/ A folder containing .txt files of the books selected sources
- led-SNAPSHOT.nt.tar.gz an archive of the Listening Experience Database in RDF/N-triples

These are the data that can be used for benchmarking knowledge extraction processes:

- **sources.csv** (columns: source, file, title, author, author_name, time)
- **experiences.csv** (columns: file, exp, excerpt, text, time, place, listening_to, environment, listener, listener_label, type, instrument, genre)
- **child.csv** includes the list of listening experiences that were marked by domain experts to be relevant to childhood

4.1 Summary

ID	documentary-evidence-benchmark
Type	Software, Data
Title	Documentary Evidence Benchmark
Credits	Alba Morales (OU) Enrico Daga (OU)
Description	This software and data provide the benchmark for the extraction of documentary evidence, taking the Listening Experience Database (LED) as a reference
Source Code	https://github.com/polifonia-project/documentary-evidence-benchmark
Documentation	README.md
Release	v1.0
Licence	Apache 2.0
Bibliography	[9] Daga, Enrico and Motta, Enrico (2019). Challenging knowledge extraction to support the curation of documentary evidence in the humanities. In: Third International Workshop on Capturing Scientific Knowledge (SciKnow). Collocated with the tenth International Conference on Knowledge Capture (K-CAP) (Garijo, Daniel; Markovic, Milan; Groth, Paul; Santana, Idafen and Belhajjame, Khalid eds.), Los Angeles, CA, USA.
Related Deliverables	Polifonia deliverable D4.3

4.2 Exploitability

The resource is targeted to evaluate advanced knowledge extraction methods tailored to the domain of historical musicology.

4.2.1 Benchmark Tasks

We briefly describe each task and refer to the relevant data.

Task 1 : retrieve documentary evidence relevant to musical experiences This task refers to automatically identify text fragments that contain an account of listening experience, from a selection of texts (in corpus/).

Input: sources.csv (source, file, title, author, author_name, time both the textual content in corpus/ and the related metadata can be used)

Target: for each file, find paragraphs that match (or overlap) with the ones in experience (text) (the other columns except file should be ignored and not used by the approach)

Task 2 : retrieve documentary evidence relevant to childhood This task is equivalent to Task 1, except that the output should match the list of experiences in child.csv

Task 3 : populate documentary evidence entities metadata This task operates on the expected output from the previous ones. Given a list of texts and related excerpts, populate the metadata describing the listening experience.

Input: sources.csv (all columns), experiences.csv (file, exp, excerpt, text)

Target: automatically derive columns in experiences.csv: place, listening_to, environment, listener, listener_label, type, instrument, genre

Task 4 : time-indexing of documentary evidence This task operates on the expected output from the previous ones. Given a list of texts and related excerpts, populate the metadata describing the listening experience.

Input: sources.csv (all columns), experiences.csv (file, exp, excerpt, text)

Target: automatically derive columns in experiences.csv: time

4.3 Benchmark construction process

The data was generated using SPARQL Anything, the following `fx` command shall be interpreted as `java -jar sparql-anything-0.7.0-SNAPSHOT.jar`.

Generate list of sources from exemplary LED entities (uncompress the `led-SNAPSHOT.nt.tar.gz` archive before executing the following).

```
fx -q queries/sources.sparql -l led-SNAPSHOT.nt -o data/sources.csv -f CSV
```

```
fx -q queries/experiences.sparql -l led-SNAPSHOT.nt -o data/sources.csv -f CSV
```

```
fx -q queries/child.sparql -l led-SNAPSHOT.nt -o data/child.csv -f CSV
```

The content of the benchmark data is summarised in Table 4.1

resource type	count
sources	25
places	277
genres	100
listeners	194
instruments	64
experiences relevant to childhood	40
experiences	1248
performances/pieces	1121

Table 4.1: Benchmark data statistics

4.4 Future work

Future work includes the development of baseline methods for each one of these tasks, by relying on state of the art techniques, and the development of innovative knowledge extraction methods to better support these tasks.

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